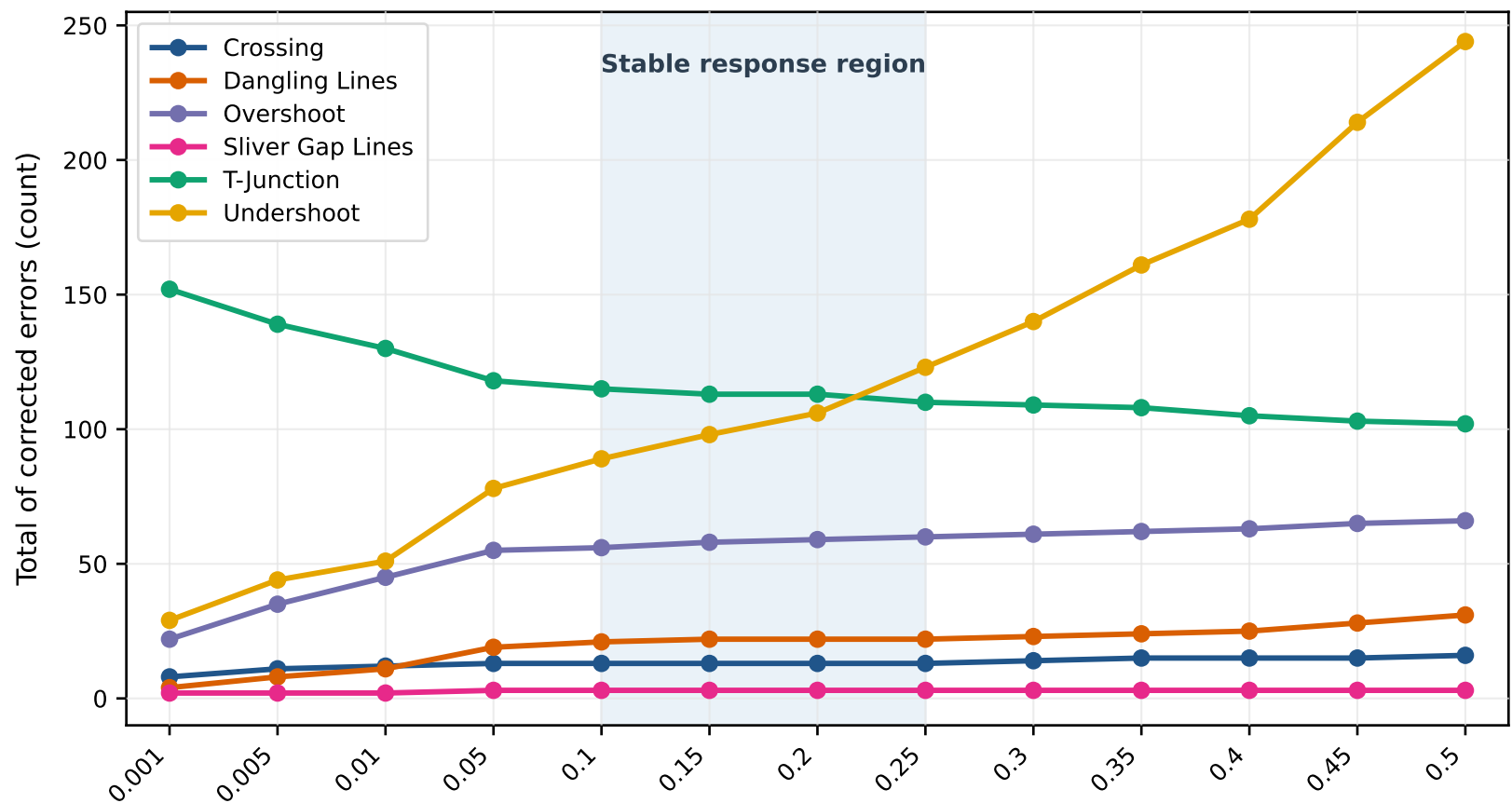


(a) Trajectory of topological defect detection across tolerance continuum

(b) Sensitivity heatmap matrix of relative deviations (%) from baseline  $\varepsilon = 0.10$ 